

The Labor Market Effects of Carbon Pricing

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Carbon Pricing and the Economy

- Cap-and-trade systems cover 26% of global greenhouse-gas emissions (ICAP 2026).
→ Several additional systems have been announced.
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⇒ Employment? Wages? Winners and losers?

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- **EU ETS** as the laboratory
- Rich **population-level** data sets from the Netherlands

The 2017 EU ETS Policy Reform



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- Average plant in sample has to surrender 154.6 permits/worker $\Rightarrow$ €3,100/worker annually.

- **Population Data:** Dutch administrative matched employer-employee dataset (CBS) + The EU Transaction Log (EUTL). Period: 2014-2020.

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- **Identification Challenge:** ETS firms are systematically larger, more profitable, and pay higher wages.
- **Matched DiD Strategy:** Match treated workers in 2016 ETS plants to control workers in non-ETS plants.
 - **Exact match:** Two-digit industry, gender, high tenure.
 - **Propensity Score:** Age, wage history ($t - 1, t - 2$), firm size, profit per worker.

$$y_{it} = \beta(\text{ETS}_i \times \text{Post}_t) + \gamma_i + \delta_t + \epsilon_{it}$$

No Adverse Average Effects

<i>Dep. Var.:</i>	<u>Log(Hr. Wage)</u>	<u>Earnings</u>	<u>Log(Wage)</u>	<u>Employed</u>
	(1)	(2)	(3)	(4)
ETS $\times$ Post	0.018 (0.016)	2,105.471 (1,482.941)	0.019 (0.017)	0.006** (0.003)
Observations	240,047	245,574	240,047	245,574
Worker FE	X	X	X	X
Year FE	X	X	X	X

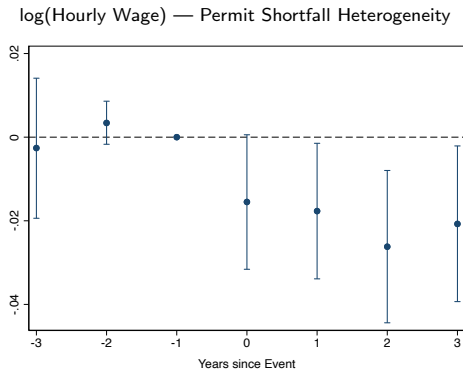
No evidence of adverse **average** wage or employment effects.

Market Design & Cash-Flow Pass-Through

- **Permit Shortfall** = Emissions – Free Permits (per worker).
- Firms with larger shortfalls experience larger cash outflows when carbon prices rise.

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- **Pass-Through to Labor**: 1 std. dev. higher permit shortfall $\Rightarrow$ **1.8% relative decline in hourly wages**.
- Robust to ETS $\times$ event-study dummies with the pre-reform average shortfall.
- **Policymakers' free-allocation choices directly affect wage distribution.**

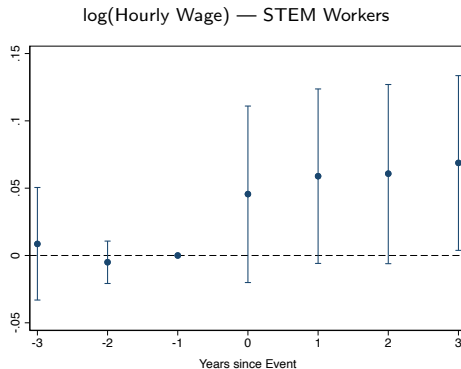


The STEM-ETS Skill Premium

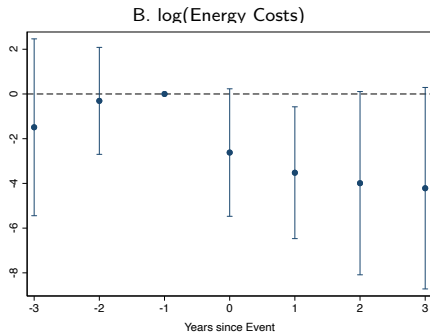
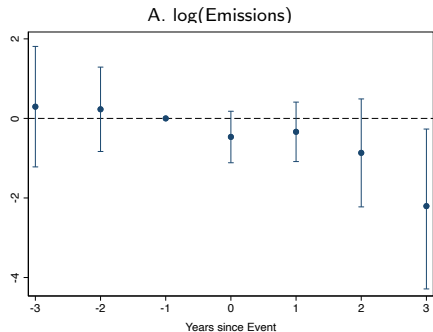
- Shock increases the revenue that workers with abatement skills generate.
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- Such workers can capture a fraction of this surplus, increasing wages.
- **STEM workers** in ETS firms earn **6.9% higher hourly wages** by $t + 3$ relative to non-ETS STEM workers.
- Robust to within-firm-year comparison.
- Stronger for workers with better outside options.



Are STEM Workers “Worth Their Price”?



- **1 std. dev. higher pre-reform plant STEM share** $\Rightarrow$ **9–10% reduction in emissions** and **12–18% reduction in energy costs**.
- Placebo test: No emission or energy reductions for business or non-STEM college degrees.
- **STEM workers have greener occupations** (ESCO definition).

Rising carbon prices don't have to harm incumbent workers on average. Yet, **market design** and **worker skills** affect who gains and who loses!

Thank You